Lecture 14: Absolutely Convergence; The Ratio and Root Test

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is absolutely convergence and able to identify if a series is converges absolutely.
- ☐ Use the Ratio Test to tell whether a series converges or diverges when possible and know the limitations of the test.
- ☐ Use the Root Test to tell whether a series converges or diverges when possible and know the limitations of the test.

Absolute Convergence

A series $\sum a_n$ converges absolutely (or is absolutely convergent) if the corresponding series of absolute value, $\sum |a_n|$ converges.

Eg 1 : $a_n = \frac{5(-1)^n}{4^n}$

$$\sum |a_n| = \sum \left| \frac{5(-1)^n}{4^n} \right| = \sum \frac{5(1)^n}{4^n}$$

$$= \frac{5}{4} + \frac{5}{16} + \frac{5}{64} + \cdots$$

(geometric series with $a = 5/4$, $r = 1/4$)

$|r| < 1$ (converges)

So we say $\sum_{n=1}^{\infty} \frac{5(-1)^n}{4^n}$ converges absolutely.

Eg 2: $a_n = \frac{(-1)^{n+1}}{n}$

$$\sum |a_n| = \sum \left| \frac{(-1)^{n+1}}{n} \right| = \sum \frac{1}{n}$$

(divergent harmonic series)

So $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ does not converge absolutely.

(we DO NOT say that a series diverges absolutely)

The Absolute Convergence Test

Absolutely Convergence Test

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

converges absolutely $\Rightarrow$ converges.

proof: We know that $-|a_n| \le a_n \le |a_n|$

Adding $|a_n|$ to the inequality,

we will get $0 \le a_n + |a_n| \le 2|a_n|$

If $\sum |a_n|$ converges, then $\sum 2|a_i|$ must also converge.

So by DCT, $\sum (a_n + |a_n|)$ also converges

Now $\sum a_n = \sum (a_n + |a_n|) - \sum |a_n|$ (converges; converges) So $\sum a_n$ converges.

The Absolute Convergence Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2}$ (convergent)

Let $a_n = (-1)^{n+1}\frac{1}{n^2}$, $\sum |a_n| = \sum \frac{1}{n^2}$

(convergent p-series)

So, $\sum_{n=1}^{\infty} a_n$ converges absolutely.

By Absolute Convergence Test,

$\sum_{n=1}^{\infty} a_n$ converges.

The Absolute Convergence Test

Example: $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$

$-1 \le \sin n \le 1$

$0 \le |\sin n| \le 1$

$0 \le \left|\frac{\sin n}{n^2}\right| \le \frac{1}{n^2}$ (convergent p-series)

Let $a_n = \frac{\sin n}{n^2}$

$\sum |a_n| = \sum \left|\frac{\sin n}{n^2}\right|$ ← converges?

By DCT, $\sum \left|\frac{\sin n}{n}\right|$ converges.

By Absolute Convergence Test,

$\sum \frac{\sin n}{n^2}$ converges.

The Absolute Convergence Test

The converse is not necessarily true. (A convergent series is not necessarily converges absolutely)

Example: $\sum_{n=1}^{\infty}(-1)^n \frac{1}{n}$ converges (we will discuss this when we talk about alternating series)

$$\sum \left|(-1)^n \frac{1}{n}\right| = \sum \frac{1}{n}$$ (divergent harmonic series)

In this case, we say the series converge conditionally. (converges but not converges absolutely)

The Ratio Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \rho.$$

Then

- the series converges absolutely if $\rho < 1$,
- the series diverges if $\rho > 1$ or ρ is infinite,
- the test is inconclusive if $\rho = 1$.

Proof

① $\rho < 1$ (the series converges absolutely) $\Rightarrow$ converges

Let r be any number between ρ and 1 $(\rho < r < 1)$

Ratio Test

By the definition of limit, $\forall \varepsilon > 0$, there exists an integer N such that for all $n \geq N$, we have

$$\left| \frac{a_{n+1}}{a_n} - \rho \right| < \varepsilon$$

Now $\rho < r \Rightarrow r - \rho > 0 \Rightarrow$ we can take $\varepsilon = r - \rho$

$$\left| \frac{a_{n+1}}{a_n} - \rho \right| < r - \rho \Rightarrow -(r-\rho) < \frac{a_{n+1}}{a_n} - \rho < r - \rho$$

$$\frac{a_{n+1}}{a_n} < r$$

$$a_{n+1} < r a_n \text{ for all } n \geq N.$$

$$\therefore \quad a_{N+1} < r a_N$$

$$a_{N+2} < r a_{N+1} < r(r a_N) = r^2 (a_N)$$

$$a_{N+3} < r a_{N+2} < r(r^2 a_N) = r^3 a_N$$

$$\vdots$$

$$a_{N+m} < r^m a_N \quad \text{for some positive integer } m.$$

$$\vdots$$

$$\text{So } \sum_{n=N+1}^{\infty} |a_n| < \sum_{n=1}^{\infty} r^n a_N$$

geometric series with $|r| < 1$ (convergent)

So by DCT, $\sum_{n=N+1}^{\infty} |a_n|$ converges

$$\Rightarrow \sum_{n=N+1}^{\infty} a_n \text{ converges} \quad \text{(By Absolutely Convergence Test)}$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n \text{ converges.}$$

② $\rho > 1$ (the series diverges)

For some integer N, we have $\frac{a_{n+1}}{a_n} > 1$ for all

$n \geq N$

$\Rightarrow a_N < a_{N+1} < a_{N+2} < \cdots$

We can see that the terms keep increasing and $\lim_{n\to\infty} a_n \neq 0$. So by the nth term test for divergence, the series diverges.

③ $\rho = 1$ (inconclusive)

Eg: $\sum \frac{1}{n}$ diverges but $\sum \frac{1}{n^2}$ converges.

$$\lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

$$\lim_{n\to\infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = 1$$

Ratio Test

Example: $\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{2^{n+1}+5}{3^{n+1}}}{\frac{2^n+5}{3^n}} = \frac{2^{n+1}+5}{3^{n+1}} \times \frac{3^n}{2^n+5} = \frac{2^{n+1}+5}{3^{n+1}\,3} \times \frac{3^n}{2^n+5}$$

$$= \frac{1}{3}\left(\frac{2^n\left(2+\frac{5}{2^n}\right)}{2^n\left(1+\frac{5}{2^n}\right)}\right)$$

This does not mean that the series converges to $\frac{2}{3}$!!

$$= \frac{1}{3}\left(\frac{2+\frac{5}{2^n}}{1+\frac{5}{2^n}}\right)$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{1}{3}\left(\frac{2+\frac{5}{2^n}}{1+\frac{5}{2^n}}\right) = \frac{2}{3} < 1 \quad \text{(converges)}$$

To look for the sum,

$$\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n} = \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n + \sum_{n=0}^{\infty} \frac{5}{3^n}$$

geometric series with $a = 1$, $r = \frac{2}{3}$

geometric series with $a = 5$, $r = \frac{1}{3}$

$$= \frac{1}{1 - \frac{2}{3}} + \frac{5}{1 - \frac{1}{3}}$$

$$= \frac{21}{2}$$

Ratio Test

Example: $\sum_{n=1}^{\infty} \frac{(2n)!}{n!n!}$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{(2(n+1))!}{(n+1)!(n+1)!}}{\frac{(2n)!}{n!n!}} = \frac{(2n+2)!}{(n+1)!(n+1)!} \times \frac{n!n!}{(2n)!}$$

$(2n+2)(2n+1)(2n)\cdots 3\cdot 2\cdot 1$

$(n+1)n!$ $\quad n+1$

$$= \frac{(2n+2)(2n+1)}{(n+1)(n+1)}$$

$$= \frac{4n^2+6n+2}{n^2+2n+1} \longrightarrow 4 > 1$$

So by Ratio Test,

The series diverges.

Ratio Test

Example: $\sum_{n=1}^{\infty} \frac{4^n n! n!}{(2n)!}$

$$\frac{a_{n+1}}{a_n} = \frac{4^{n+1}\,(n+1)!\,(n+1)!}{(2(n+1))!} \times \frac{(2n)!}{4^n\, n!\, n!}$$

$(2n+2)(2n+1)$

$$= \frac{4(n+1)(n+1)}{(2n+2)(2n+1)} \longrightarrow 1 \quad \text{(inconclusive)}$$

So we have to use other test.

It is easy to check that $a_{n+1} > a_n > 0$ for all n.

So a_n does not converge to 0. So by n^{th} term test for divergence, the series diverges.

Ratio Test

Example: $\sum_{n=0}^{\infty} \frac{n^2 2^n}{3^n}$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2 \, 2^{n+1}}{3^{n+1}} \times \frac{3^n}{n^2 2^n}$$

$$= \frac{2}{3} \frac{(n+1)^2}{n^2} \longrightarrow \frac{2}{3} < 1$$

So by the Ratio Test, $\sum_{n=0}^{\infty} \frac{n^2 2^n}{3^n}$ converges.

Root Test

Consider the series with the terms

$$a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$$

What happens if we apply ratio test to the series?

If n is odd, $\dfrac{a_{n+1}}{a_n} = \dfrac{1/2^{n+1}}{n/2^n} = \dfrac{1}{2n}$

If n is even, $\dfrac{a_{n+1}}{a_n} = \dfrac{(n+1)/2^{n+1}}{1/2^n} = \dfrac{n+1}{2}$

$\lim_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right|$ does not exist, so ratio test cannot be applied.

Root Test

The Root Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \rho.$$

Then

- the series converges absolutely if $\rho < 1$,
- the series diverges if $\rho > 1$,
- the test is inconclusive if $\rho = 1$.

Proof ① $\rho < 1$

$\varepsilon + \rho < 1$

$\rho < 1 \Rightarrow 1 - \rho > 0$, so we can take $0 < \varepsilon < 1 - \rho$

and there is an integer N such that whenever $n \geq N$

$$\left| \sqrt[n]{|a_n|} - \rho \right| < \varepsilon$$

$$-(1+\rho) < -\varepsilon < \sqrt[n]{|a_n|} - \rho < \varepsilon$$

$$-1+\rho < -\varepsilon$$

$$-1 < -\varepsilon - \rho$$

$$= -(\varepsilon - \rho)$$

$$\sqrt[n]{|a_n|} < \varepsilon + \rho$$

$$|a_n| < (\varepsilon+\rho)^n$$

Let $\{c_n\}_{n\geq N} = \{(\varepsilon+\rho)^n\}_{n\geq N} = \{(\varepsilon+\rho)^N, (\varepsilon+\rho)^{N+1}, (\varepsilon+\rho)^{N+2}, \ldots\}$

Then $\sum_{n=N}^{\infty} c_n$ will be geometric series with $r = \varepsilon+\rho < 1$

$|r| < 1$

(convergent)

By DCT $\sum |a_n|$ converges $\Rightarrow \sum a_n$ converges.

② $\rho > 1$. There must be an integer N such that

$\sqrt[n]{c_n} > 1$ for all $n \geq N \Rightarrow c_n > 1 \not\to 0$ for all $n \geq N$.

$\Rightarrow \sum a_n$ diverges by the nth term test for divergence.

(B) $\rho = 1$

$\sum \frac{1}{n}$ diverges but $\sum \frac{1}{n^2}$ converges.

$\lim_{n\to\infty} \sqrt[n]{\frac{1}{n}} = 1$ $\qquad$ $\lim_{n\to\infty} \sqrt[n]{\frac{1}{n^2}} = 1$

Exercise: Check these.

Root Test

Back to the series with the terms

$$a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$$

$$\sqrt[n]{|a_n|} = \begin{cases} \sqrt[n]{\left|\frac{n}{2^n}\right|}, & n \text{ is odd} \\ \sqrt[n]{\left|\frac{1}{2^n}\right|}, & n \text{ is even} \end{cases}$$

$$= \begin{cases} \frac{\sqrt[n]{n}}{2} \to \frac{1}{2}, & n \text{ is odd} \\ \frac{1}{2} \to \frac{1}{2}, & n \text{ is even.} \end{cases}$$

$\lim_{n\to\infty} \sqrt[n]{n} = 1$

So $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \frac{1}{2} < 1$. By the root test, the series converges absolutely and hence it must be convergent.

Root Test

Example: $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

Let $a_n = \frac{n^2}{2^n}$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{n^2}{2^n}\right|}$$

$$= \lim_{n\to\infty} \sqrt[n]{\frac{n^2}{2^n}}$$

$$= \lim_{n\to\infty} \frac{n^{2/n}}{2}$$

$$= \frac{1}{2} \lim_{n\to\infty} \underbrace{\left(n^{1/n}\right)^2}_{\to 1} = \frac{1}{2} < 1$$

By the Root Test, the series converges absolutely.

$\Rightarrow$ the series converges.

Root Test

Example: $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$ Let $a_n = \frac{2^n}{n^3}$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{2^n}{n^3}\right|}$$

$$= \lim_{n\to\infty} \frac{2}{n^{3/n}}$$

$$= \frac{2}{\left(\lim_{n\to\infty} \sqrt[n]{n}\right)^3} = 2 > 1$$

So by the Root Test, the series diverges.

Root Test

Example: $\sum_{n=1}^{\infty}\left(\frac{1}{1+n}\right)^n$

Let $a_n = \left(\frac{1}{1+n}\right)^n$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{1}{1+n}\right)^n\right|}$$

$$= \lim_{n\to\infty} \frac{1}{1+n} = 0 < 1$$

By the Root Test, the series converges absolutely

$\Rightarrow$ the series converges.

Root Test

Example: $\sum_{n=1}^{\infty}(1-\frac{1}{n})^{n^2}$

Let $a_n = \left(1-\frac{1}{n}\right)^{n^2}$

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\left|\left(1-\frac{1}{n}\right)^{n^2}\right|}$$

$$= \lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{\frac{n\times n}{n}}$$

$$= \lim_{n\to\infty}\left(1+\left(\frac{-1}{n}\right)\right)^{n}$$

$$= e^{-1} < 1$$

By the Root Test, the series converges absolutely.

$\Rightarrow$ the series converges